

Richard Stiskalek^{ID}, Curriculum Vitae

PROFILE	Affiliation: University of Oxford Email: richard.stiskalek@physics.ox.ac.uk NASA ADS: 16 first-author, 15 co-author papers; <i>h</i>-index = 10; citations = 736 Website: richard-sti.github.io I am a Hintze Prize Fellow in Astrophysics (Cosmology) and specialize in Bayesian field-level inference and digital twins of the local Universe to bridge observations and theory. My research combines constrained cosmological simulations, semi-analytic galaxy formation models, simulation-based inference, and machine learning to test galaxy physics, map large-scale flows, and deliver precision measurements of cosmological parameters.
EDUCATION	University of Oxford (Balliol College), DPhil in Astrophysics 2022 - 2026 Thesis: “Testing the local Universe with constrained cosmological simulations” Supervised by <i>Harry Desmond</i>, <i>Julien Devriendt</i>, and <i>Adrianne Slyz</i> Ludwig-Maximilians-Universität München, M.Sc. in Physics 2020 – 2022 Thesis: “Strong-field lensing of gravitational waves” (partially with <i>Albert Einstein Institute Potsdam</i>) Supervised by <i>Miguel Zumalacárregui</i>, <i>Marius A. Oancea</i>, and <i>Jochen Weller</i> Hong Kong University of Science and Technology 2017 – 2018 Undergraduate Exchange Programme in Physics University of Glasgow, B.Sc. in Physics with Astrophysics 2016 – 2020 Thesis: “Gravitational-wave cosmology” Supervised by <i>Martin Hendry</i>
ACADEMIC APPOINTMENTS	University of Oxford, Hintze Centre for Astrophysical Surveys 2026 – 2029 Hintze Prize Fellow in Astrophysics (Cosmology) Flatiron Institute, Center for Computational Astrophysics 2025 Pre-Doctoral Fellow (with <i>Shy Genel</i> and <i>Lucia A. Perez</i>) Institut d’astrophysique de Paris 2024 Long Term Attachment (with <i>Guilhem Lavaux</i>) Max Planck Institute for Gravitational Physics, Observational Relativity and Cosmology 2020 Summer Research Intern (with <i>Collin Capano</i>) University of Oxford, Sub-department of Astrophysics 2019 Summer Research Intern (with <i>Harry Desmond</i>) University of Glasgow, Institute for Gravitational Research 2018 Summer Research Intern (with <i>John Veitch</i> and <i>Chris Messenger</i>)
COLLABORATION MEMBERSHIPS	Learning the Universe 2025 – Simons Foundation collaboration developing next-generation cosmological simulations, machine learning methods, and advanced Bayesian inference for large-scale surveys <i>Contributions:</i> Training set generation and BORG team Aquila Consortium 2022 – International collaboration developing Bayesian field-level inference techniques to reconstruct initial conditions and study large-scale structure formation <i>Contributions:</i> Leading projects on data generation, benchmarking, large-scale flows and cosmological parameters
TEACHING, MENTORING & SERVICE	Student Supervision <i>Noah Pierce</i> (BSc, Southampton; with H. Desmond & T. Yasin) 2025 – present “Field-Level Inference of the Galaxy Luminosity Function” <i>Joshua Darne</i> (MPhys, Oxford; with T. Yasin & H. Desmond) 2024 “Radial acceleration relation in the NewHorizon hydrodynamical simulation”

<i>Fedir Boreiko</i> (BSc, Manchester; with T. Yasin & H. Desmond)	2024
“The correlation between light and dark matter across cosmic time”	
<i>Enoch Ko</i> (BSc, Warwick; with T. Yasin & H. Desmond)	2024
“Dark matter and galaxy dynamics: enduring puzzles” [arXiv:2508.03569]	
<i>Catherine Gallagher</i> (MPhys, Oxford; with T. Yasin & H. Desmond)	2023 – 2024
“The influence of cosmic environment on galaxy properties” [arXiv:2503.14732]	
<i>James Harvey</i> (BSc, Oxford; with T. Yasin & H. Desmond)	2023 – 2024
“Machine learning the time of last major merger from spectroscopic data”	

Teaching Experience

Tutor, <i>MPhys C1 Astrophysics</i> , University of Oxford	2023 – 2024
Graduate tutorials in cosmology and stellar astrophysics	
Mentor, <i>Lumiere Education</i>	2023 – present
Supervision of senior high school students carrying out independent research projects	
Demonstrator, <i>3rd-Year Practical Course</i> , University of Oxford	2023
Computational astrophysics practical sessions	

Academic Service

Referee for <i>A&A</i> , <i>ApJ</i> , <i>MNRAS</i> , <i>PNAS</i> , and <i>PRD</i>	2022 – present
Reviewed 15+ manuscripts for leading journals	
Coordinator, Aquila Consortium Monthly Telecon	2023 – present
Organiser, “Simulator’s Coffee”, University of Oxford	2023 – 2024
Local organiser, Aquila Consortium Oxford Meeting	2023
Lead organiser, “Middle of Scotland Science Festival”	2018

SELECTED	Snell Exhibition, Balliol College	2022 - 2026
AWARDS AND	STFC PhD Funding, Science and Technology Facilities Council	2022 - 2026
SCHOLARSHIPS	DAAD Study Scholarship, German Academic Exchange Service	2021 - 2022
	Kerr Bursary, University of Glasgow	2020
	Lang Scholarship, University of Glasgow	2019
	Undergraduate Summer Bursary, Royal Astronomical Society	2018
	Dean’s List, Hong Kong University of Science and Technology	2018
	Astronomy 1 Prize, University of Glasgow	2017
	Matthew A Muir Bursary, University of Glasgow	2017
	South East Asia Study Abroad Scholarship, University of Glasgow	2017 - 2018

SKILLS	Programming: Python, Julia, C/C++, Fortran; high-performance computing (MPI, GPU acceleration) Statistics & Machine Learning: Bayesian inference (NumPyro/JAX, PyMC, Stan), simulation-based inference, density estimation; deep learning (MLPs, CNNs, GNNs, normalizing flows) with JAX, TensorFlow, PyTorch; tree-based methods (random forests, gradient boosting) with scikit-learn Astrophysics Software: RAMSES, AREPO, Gadget, Rockstar, DisPerSe; <i>N</i>-body/hydrodynamical simulation analysis Languages: English (fluent), Czech (native), Slovak (fluent), French (intermediate), German (beginner)
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PUBLICATIONS	First-Author – Submitted [1] “Forward-modelling Milky Way Cepheids: selection effects and physical priors in the Gaia–HST calibration”; R. Stiskalek, A. Riess, H. Desmond, G. Lavaux, D. Scolnic [arXiv:2603.09880] [2] “S_8 from Tully–Fisher, fundamental plane, and supernova distances agree with <i>Planck</i>”; R. Stiskalek [arXiv:2509.20235] First-Author – Accepted
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- [3] “Validating Digital Twins of the Local Universe with the Thermal Sunyaev-Zel’dovich Signal”; **R. Stiskalek**, H. Desmond [Open Journal of Astrophysics, arXiv:2601.15935]
- [4] “Revisiting the Great Attractor: The Local Group’s streamline trajectory, cosmic velocity and dynamical fate”; **R. Stiskalek**, H. Desmond, S. McAlpine, G. Lavaux, J. Jasche, M. J. Hudson [Open Journal of Astrophysics, arXiv:2601.08524]
- [5] “1.8 per cent measurement of H_0 from Cepheids alone”; **R. Stiskalek**, H. Desmond, E. Tsaprazi, A. Heavens, G. Lavaux, S. McAlpine, J. Jasche [MNRAS, arXiv:2509.09665]
- [6] “No evidence for local H_0 anisotropy from Tully–Fisher or supernova distances”; **R. Stiskalek**, H. Desmond, G. Lavaux [MNRAS 546:staf2048, arXiv:2509.14997]
- [7] “The Velocity Field Olympics: Assessing velocity field reconstructions with direct distance tracers”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz, G. Lavaux, M. Hudson, D. Bartlett, H. Courtois [MNRAS 545:staf1960, arXiv:2502.00121]
- [8] “Testing the local supervoid solution to the Hubble tension with direct distance tracers”; **R. Stiskalek**, H. Desmond, I. Banik [MNRAS 543:1556, arXiv:2506.10518]
- [9] “Symmetry in Hyper Suprime-Cam galaxy spin directions”; **R. Stiskalek**, H. Desmond [Res. Notes AAS 8 281, arXiv:2410.18884]
- [10] “Evaluating the variance of individual halo properties in constrained cosmological simulations”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz [MNRAS 534:3120, arXiv:2310.20672]
- [11] “Probing general relativistic spin-orbit coupling with gravitational waves from hierarchical triple systems”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [MNRAS 535:L1, arXiv:2307.01903]
- [12] “On the fundamentality of the radial acceleration relation for late-type galaxy dynamics”; **R. Stiskalek**, H. Desmond [MNRAS 525:6130, arXiv:2305.19978]
- [13] “Frequency- and polarization-dependent lensing of gravitational waves in strong gravitational fields”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [Phys. Rev. D 109, 124045, arXiv:2209.06459]
- [14] “The scatter in the galaxy–halo connection: a machine learning analysis”; **R. Stiskalek**, D. J. Bartlett, H. Desmond, D. Anbajagane [MNRAS 514:4026, arXiv:2202.14006]
- [15] “The dependence of subhalo abundance matching on galaxy photometry and selection criteria”; **R. Stiskalek**, H. Desmond, T. Holvey, M. G. Jones [MNRAS 506:3205, arXiv:2101.02765]
- [16] “Are stellar-mass binary black hole mergers isotropically distributed?”; **R. Stiskalek**, J. Veitch, C. Messenger [MNRAS 501:970, arXiv:2003.02919]

Co-Author – Submitted

- [1] “Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation”; V. Pandya, . . . , **R. Stiskalek**, . . . [arXiv:2604.06318]
- [2] “Parameterizing Dark Energy at the density level: A two-parameter alternative to CPL”; G. Montefalcone, **R. Stiskalek** [arXiv:2603.25735]
- [3] “Testing cosmic anisotropy with cluster scaling relations”; T. Yasin, **R. Stiskalek**, H. Desmond, S. von Hausegger, P. G. Ferreira [arXiv:2602.06007]
- [4] “Constraining cosmological simulations with peculiar velocities: a forward-modelling approach”; A. Valade, N. Libeskind, D. Pomarède, **R. Stiskalek**, Y. Hoffman, S. Gottlöber, R. B. Tully [arXiv:2602.03699]
- [5] “Testing subhalo abundance matching with galaxy kinematics”; F. Boreiko, T. Yasin, H. Desmond, **R. Stiskalek**, M. Jarvis [arXiv:2601.07799]
- [6] “The subtle statistics of the distance ladder: On the distance prior and selection effects”; H. Desmond, **R. Stiskalek**, J. A. Najera, I. Banik [arXiv:2511.03394]
- [7] “MEGATRON: Reproducing the Diversity of High-Redshift Galaxy Spectra with Cosmo-

logical Radiation Hydrodynamics Simulations”; H. Katz, . . . , **R. Stiskalek**, . . . [arXiv:2510.05201]

- [8] “MEGATRON: how the first stars create an iron metallicity plateau in the smallest dwarf galaxies”; M. Rey, . . . , **R. Stiskalek**, . . . [arXiv:2510.05232]
- [9] “MEGATRON: the impact of non-equilibrium effects and local radiation fields on the circumgalactic medium at cosmic noon”; C. Cadiou, . . . , **R. Stiskalek**, . . . [arXiv:2510.05667]

Co-Author – Accepted

- [10] “The galaxy-environment connection revealed by constrained simulations”; C. Gallagher, T. Yasin, **R. Stiskalek**, H. Desmond, M. Jarvis [MNRAS:stag108, arXiv:2503.14732]
- [11] “Renzo’s rule revisited: A statistical study of galaxies baryon–dark matter coupling”; E. Ko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [MNRAS 544:4288, arXiv:2508.03569]
- [12] “CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning”; N. Huang, **R. Stiskalek**, J.-Y. Lee, A. E. Bayer, C. C. Margosian, C. K. Jespersen, L. A. Perez, L. K. Saul, F. Villaescusa-Navarro [NeurIPS 2025, arXiv:2507.03707]
- [13] “The Manticore Project I: a digital twin of our cosmic neighbourhood from Bayesian field-level analysis”; S. McAlpine, J. Jasche, M. Ata, G. Lavaux, **R. Stiskalek**, C. S. Frenk, A. Jenkins [MNRAS 540:716, arXiv:2505.10682]
- [14] “The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics”; Eleonora Di Valentino, . . . , **R. Stiskalek**, . . . [Physics of the Dark Universe, Volume 49, 2025, 101965, arXiv:2504.01669]
- [15] “Inferring the Ionizing Photon Contributions of High-Redshift Galaxies to Reionization with JWST NIRCам Photometry”; N. Choustikov, **R. Stiskalek**, A. Saxena, H. Katz, J. Devriendt, A. Slyz [MNRAS 537:2273, arXiv:2405.09720]

INVITED TALKS	<i>Supernova-free local ladders for H_0</i>	2026
	GDR CoPhy Episode 4, Clermont-Ferrand, France	
	<i>Bridging Machine Learning and Cosmology at Scale</i>	2026
	PV Reconstruction online meeting	
	<i>Is the Hubble tension easing?</i>	
	University of Oxford	
	<i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i>	
	CAMELS + Baryon Pasting Workshop, University of Osaka, Japan	
	<i>Velocity Field Olympics</i>	
	Leibniz Institute for Astrophysics Potsdam, Germany	
	<i>Forward modelling of peculiar velocities and large-scale structure</i>	2025
	Institut d’Astrophysique de Paris	
	<i>Pinning down the local Universe with digital twins</i>	
	University of Cambridge	
	<i>Pinning down the local Universe with digital twins</i>	
	Yale University	
	<i>Do SH0ES Cepheids prefer the Planck Hubble constant?</i>	
	National Astronomy Meeting, Durham	
	<i>Towards accurate models and tests of the local Universe</i>	
	Columbia University	
	<i>Towards accurate models and tests of the local Universe</i>	
	Princeton University	
	<i>Towards accurate models and tests of the local Universe</i>	
	Center for Computational Astrophysics, Flatiron Institute	
	<i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i>	
	Center for Computational Astrophysics, Flatiron Institute	
	<i>Velocity Field Olympics</i>	
	Center for Computational Astrophysics, Flatiron Institute	
	<i>Velocity Field Olympics</i>	
	Cosmic Flows 2025, Brisbane	

<i>Velocity Field Olympics</i> University of Portsmouth	2024
<i>Search for the optimal dark matter halo density profile</i> University of Oxford	2023
<i>Is the radial acceleration relation a fundamental correlation?</i> University of Oxford	
<i>Frequency and polarization dependent propagation of gravitational waves</i> University of Glasgow	2022
<i>Frequency and polarization dependent propagation of gravitational waves</i> LMU Munich	
<i>Frequency and polarization dependent propagation of gravitational waves</i> Max Planck Institute for Gravitational Physics, Potsdam	
<i>The scatter in the galaxy–halo connection</i> Baryon Pasters Collaboration meeting	
<i>The scatter in the galaxy–halo connection</i> LMU Munich	2021
<i>Reversible-jump MCMC in gravitational-wave astronomy</i> Max Planck Institute for Gravitational Physics, Hannover	2020
<i>Are binary-black hole mergers isotropically distributed?</i> LIGO Scientific Collaboration Data Analysis telecon	
<i>The relation between galaxies and dark matter halos</i> University of Oxford	2019